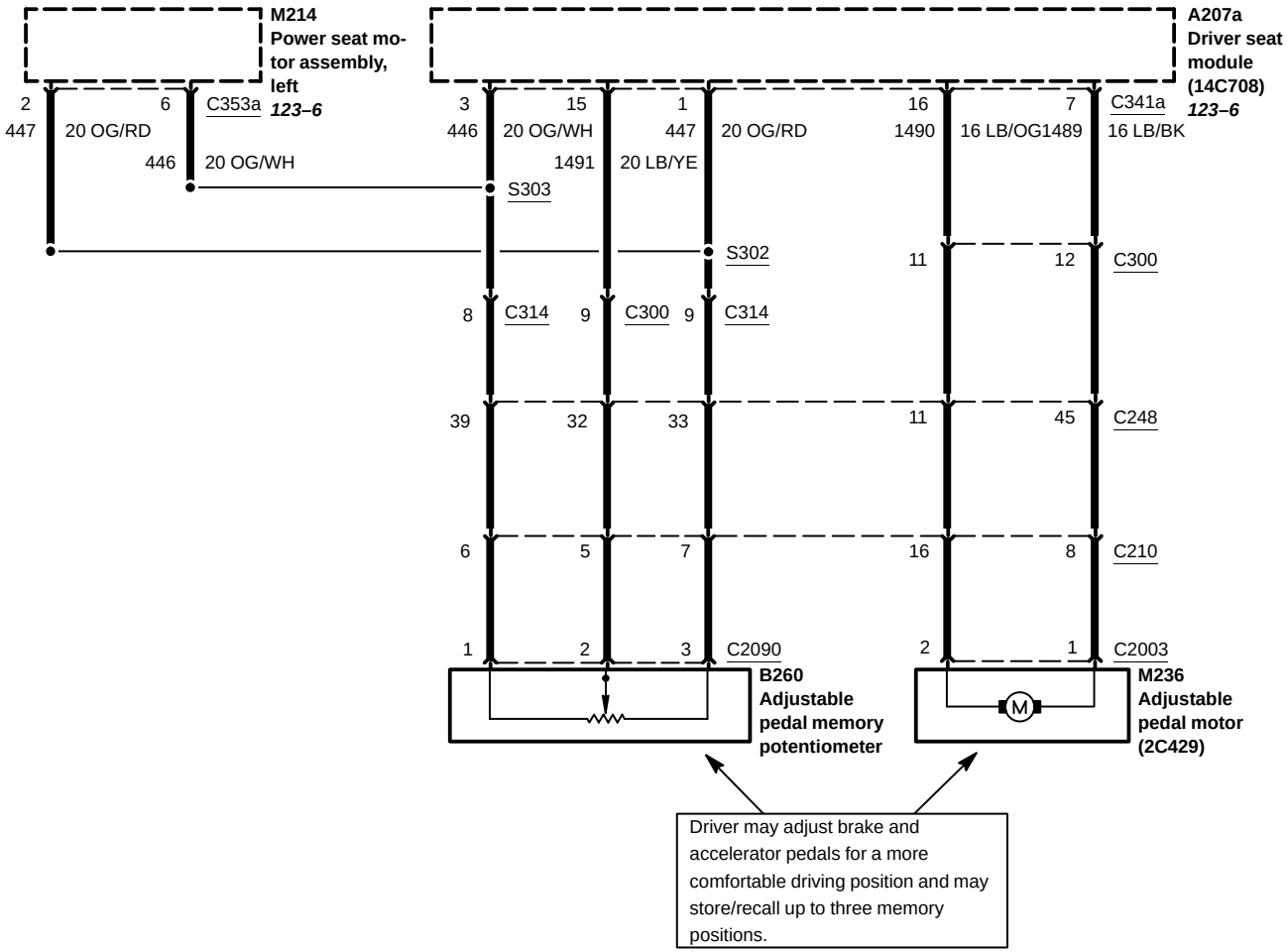Early build, with memory